

Exploring effective work unit knowledge management (KM): roles of network, task, and KM strategies

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Abstract

Purpose – This paper aims to explore the potential contribution of inter-unit network structure and intra-unit task environment to the overall knowledge management (KM) effectiveness of an organization through different KM strategies.

Design/methodology/approach – The survey method was used to empirically test the hypotheses. The authors administered two survey forms: a member-survey form that obtained responses from members of each work unit and a leader-survey form issued only to the leader of each unit to obtain his/her perception on inter-unit communications.

Findings – The findings indicate that both a unit's boundary-spanning role and its extensive interaction with other units help promote the unit's overall KM effectiveness. In addition, the authors found that task variety was significantly related to the use of both personalization and codification of KM strategies in the unit, and that a unit's KM strategy played a mediating role between intra-unit task characteristics and KM effectiveness.

Originality/value – Few studies have attempted to address what makes some work units within an organization more effective in their use of knowledge than others. This study provides an understanding of the influence of inter-unit social network structure on KM effectiveness, as well as the influence of intra-unit task environment and KM strategies on KM effectiveness. The findings suggest that inter-unit knowledge exchange and a unit's brokering role can be a vital component of a work unit's KM strategy, and that for successful KM in a work unit, an organization should not only establish appropriate inter-unit relationship but also develop the unit's own KM strategies.

Keywords Network structure, Task characteristics, Knowledge management strategy, Knowledge management effectiveness

Paper type Research paper

1. Introduction

Many organizations have initiated knowledge management (KM) practices to improve the effectiveness of knowledge creation, transfer and utilization, as knowledge has been acknowledged as a sustainable basis for competitive advantage in organizations (Zack, 1999; Teece, 2000; Wong *et al.*, 2015; Downes and Marchant, 2016). Much attention has been given to KM practices (Alavi and Leidner, 2001; Brown and Duguid, 2001; Hansen *et al.*, 1999; Kim *et al.*, 2014), as well as how knowledge is used to create value within organizations (Gold *et al.*, 2001; Lee and Choi, 2003; Becerra-Fernandez and Sabherwal, 2014). While prior studies have deepened our understanding of successful KM at the organizational level (Hansen *et al.*, 1999; Wasko and Faraj, 2005; Zack *et al.*, 2009; Krylova *et al.*, 2016), less attention has been directed at KM practices at the work unit level. Considering that a typical employee spends most of his/her work days interacting with other members within the same work unit, there is a vital need to integrate knowledge from various work units to effectively manage the process of knowledge exchange within an organization. Such focus on a work unit level is also expected to expand our understanding of the unit's boundary-spanning role and its extensive interaction

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with other units for overall KM success. Recognizing the significance of this work unit-level perspective, few studies have attempted to examine why some work units within the organization are more effective in their use of knowledge than others (Sabherwal and Becerra-Fernandez, 2003; Cummings, 2004). More empirical work and theoretical considerations that focus on the unit of analysis on the work unit level are needed to expand insights on KM effectiveness (Bosch-Sijtsema *et al.*, 2009).

This study examines KM effectiveness at the level of the work unit, at two different levels: the inter-unit level and the intra-unit level. To address KM effectiveness at the inter-unit level, the first research question is proposed:

RQ1. How does a work unit's network structure affect the unit's KM effectiveness?

To answer this question, our research uses the social network perspective (Burt, 1992; Coleman, 1988; Hollenbeck and Jamieson, 2015). This literature posits that network structures facilitate exchange relationships that afford opportunities and assets that would otherwise not be available. Further, network characteristics (i.e. the manner in which a work unit exchanges information with other units, such as centrality, tie strength and structural holes) are positively related to knowledge transfer across organizational subunits (Hansen, 1999, 2002; Huang *et al.*, 2010; Tortoriello, 2015), group/team effectiveness (Oh *et al.*, 2004; Mehta and Mehta, 2017) and innovation (Ahuja, 2000; Tsai, 2001; López-Nicolás and Meroño-Cerdán, 2011; Jiménez-Jiménez *et al.*, 2014). Therefore, to the extent that an organization takes these network characteristics into consideration, it may be able to improve KM effectiveness at the work unit level and ultimately enhance its overall KM effectiveness.

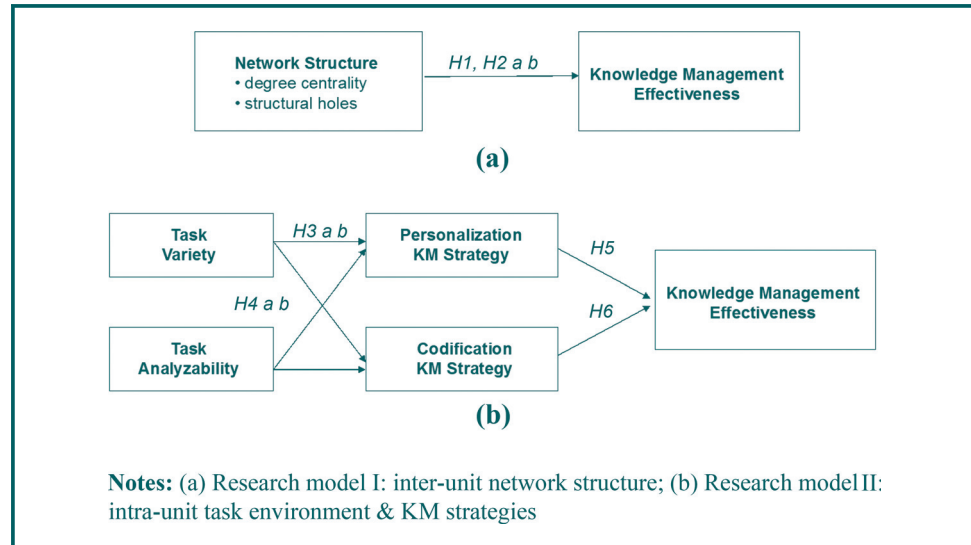
To explore KM effectiveness at the intra-unit level, our research applies two theoretical perspectives, the task environment perspective (Perrow, 1967; Wipawayangkool and Teng, 2016) and KM strategy literature (Hansen *et al.*, 1999; Choi *et al.*, 2008; Dayan *et al.*, 2017). While work units within an organization need quality information to accomplish their designated tasks (Galbraith, 1973; Premkumar *et al.*, 2005; Ben-Menahem *et al.*, 2016), these various units also differ in their information processing requirements; such requirements are characterized by the extent to which the task contains variety and is analyzable (Perrow, 1967; Daft and Lengel, 1986; Chae *et al.*, 2015). Further, the form and nature of the knowledge that is being processed within those various units may vary. Some units prefer to deal with codified and explicitly represented forms of knowledge, whereas others may rely on the tacit knowledge of personnel to solve problems. Accordingly, researchers have identified two different KM strategies: codification and personalization (Hansen *et al.*, 1999; Kim *et al.*, 2014). Because most KM activities occur daily within individual work units, it is appropriate to study KM effectiveness at the individual work unit, considering the differences of an individual unit's task environments and its KM strategies. In turn, the KM activities of individual work units may be optimized to account for differences in task characteristics. Thus, the second research question we will address in the study is:

RQ2. Given the task environment within a work unit, how do different KM strategies affect its KM effectiveness?

The paper is organized as follows. Section 2 presents two research models corresponding to the two research questions identified above. This is followed by a list of hypotheses based upon the research model. Next, Section 3 describes the research methodology and the research variables. Section 4 presents the study results and a discussion of research findings and concludes the paper with a discussion of the limitations and the implications of the study.

2. Research models and hypotheses

We developed two research models to address the two research questions. As shown in Figure 1, the first model attempts to study the influence of inter-unit network

Figure 1 Research models and hypotheses

structure on KM effectiveness. The overarching theoretical basis for this model is the social network perspective, which has been successfully applied to explain KM process and effectiveness (Song *et al.*, 2007; Kankanhalli *et al.*, 2011; Tortoriello, 2015). The social network literature widely discusses the impact that social networks have on social capital (Appendix 2), including KM phenomena (e.g. knowledge sharing and utilization) (Hansen, 2002; Lin and Lo, 2015). Researchers (Reagans and McEvily, 2003; Nakauchi *et al.*, 2017) have postulated the impact of network characteristics on organizational effectiveness by mainly focusing on knowledge sharing or transfer. We, however, believe that network characteristics significantly relate to all KM activities – knowledge creation, acquisition, transfer and utilization – across organizational subunits. Thus, our research model proposes that the network structure represents a vital social capital (Coleman, 1988; Hollenbeck and Jamieson, 2015) and posits that the presence of a social component can enhance overall KM effectiveness.

For the second model in Figure 1, we seek to examine the extent to which task environmental factors within individual units may influence KM effectiveness through different KM strategies. From the task environment perspective (Perrow, 1967), two underlying task characteristics – task variety and task analyzability (Daft and Lengel, 1986; Chae *et al.*, 2015; Lee *et al.*, 2015) – play critical roles in processing information within and across work units. KM strategies strongly relate to the way of managing knowledge to carry out day-to-day tasks more effectively. In addition, KM strategies help individual units to manage and organize available knowledge resources by interacting with existing organizational infrastructure (Yang, 2010). Further, work units have demonstrated a capacity to leverage and utilize knowledge in a relevant and timely manner in concordance with their KM strategies (Yang, 2010; Kim *et al.*, 2014). Accordingly, in the second model, we have examined KM effectiveness for individual units with key variables: task variety, task analyzability and KM strategies.

2.1 Effectiveness as outcome measure

For the study of KM, like other management phenomena, researchers need to identify a suitable success measure. Previous KM studies have used organization performance (Lee and Choi, 2003), organizational effectiveness (Gold *et al.*, 2001; Zheng *et al.*, 2010), KM performance (Kim *et al.*, 2014; Lee and Wong, 2015; Chang *et al.*, 2012) and KM

effectiveness (Chou *et al.*, 2005; Lin, 2007; Rusly *et al.*, 2012; Sabherwal and Becerra-Fernandez, 2003; Song *et al.*, 2007; Tseng and Fan, 2011) as success measures (Table I). KM effectiveness for any entity involves an assessment of “whether the entity receives and understands the knowledge needed to perform its tasks” (Sabherwal and Becerra-Fernandez, 2003, p. 227). To access perceived KM effectiveness, previous studies (Chou *et al.*, 2005; Michaelis *et al.*, 2015; Sabherwal and Becerra-Fernandez, 2003; Song *et al.*, 2007) measured general KM satisfaction. In addition, prior studies have extensively used overall satisfaction with available knowledge and utility of KM as proxies to assess KM effectiveness (Tseng and Fan, 2011; Rusly *et al.*, 2012).

According to Alavi and Leinder (2001), KM effectiveness may be conceptualized by focusing on the extent of success in the KM process: the creation, storage/retrieval, transfer and application of knowledge for task performance in the organization (Gupta and Govindarajan, 2000; Lin, 2007). In this study, we attempt to conceptualize KM effectiveness as KM satisfaction with the process of knowledge creation, acquisition, transfer and application by following previous studies (Sabherwal and Becerra-Fernandez, 2003; Lin, 2007; Tseng and Fan, 2011).

2.2 Inter-unit network structure and knowledge management effectiveness

The impact of social networks on social capital and KM activities has been widely discussed in the literature (Burt, 2000; Brass *et al.*, 2004; Cantner and Joel, 2011; Jiménez-Jiménez *et al.*, 2014). We focus on investigating the effect that the network position of intra-organizational units has on their KM effectiveness. A unit may enjoy informational benefits by forming a network relationship with other units that have complementary resources. Previous studies have shown that these resource exchange networks lead to benefits such as enhancing innovation (Tsai and Ghoshal, 1998; Tsai, 2001; Schilling and Phelps, 2007; Chen *et al.*, 2008; Tortoriello, 2015), facilitating knowledge transfer (Tsai, 2002; Reagans and McEvily, 2003; Nakauchi *et al.*, 2017) and improving performance (Oh *et al.*, 2004; Zaheer and Bell, 2005; Reinholt *et al.*, 2011). We consider two network characteristics that may influence KM effectiveness of intra-organizational units: *degree centrality* and *structural holes* (Burt, 1997; Hansen, 2002; Nerkar and Paruchuri, 2005).

Centrality exemplifies a work unit's power relative to other units (Freeman, 1979; Ibarra, 1993). Thus, the degree of centrality in a work unit's network positions reflects the extent to which the unit is involved in resource exchange. Units that are more central in a resource exchange network are likely to engage in forging new intra-organizational linkages that can lead to benefits such as knowledge and resource sharing, mutual learning and access to complementary skills (Tsai, 2000; Anderson, 2008; Björk and Magnusson, 2009; Reinholt *et al.*, 2011; Lin and Lo, 2015). In addition, a central unit is capable of applying other units' knowledge or practices to deal with complex challenges (Tsai, 2001). Further, the degree of centrality may be associated with Uzzi (1997)'s notion of *embeddedness*, which implies that the focal unit regularly shares resources, exchanges knowledge and collaborates with other units to solve problems. Units with a higher degree of centrality are able to improve access to the quality of information, which in turn enhances its performance (Hansen, 2002; Tsai, 2001; Durmuşoğlu, 2013). Therefore:

- H1. The centrality of a work unit's network position is positively associated with the work unit's KM effectiveness.

Another prominent characteristic of a unit's network position is the structural hole. Structural holes refer to the extent to which a unit in a network has access to unconnected non-redundant contacts (Burt, 1992; Zaheer and Bell, 2005). In the context of intra-organizational units, a structural hole would be a buffer between units that are themselves not connected. Structural holes provide an opportunity for a work unit to access diverse information and knowledge, leading to richer content (Hargadon and Sutton, 1997; Nerkar

Table I Review of KM measures

Previous studies	Focused KM processes	Independent variables	Mediators or moderators	Dependent variable	Measured items for dependent variable
Kim <i>et al.</i> (2014)	Accumulating/sourcing knowledge	KM strategy	Organizational information systems maturity	KM performance	Problem-solving, communication, employee abilities, product value, customer satisfaction
Lee and Wong (2015)	Acquisition, creation, application, storing, transferring and sharing	N/A	N/A	KM performance	Knowledge resources, KM processes, KM factors (culture, leadership, organizational infrastructure, strategy, resource, HRM)
Chang <i>et al.</i> (2012)	Socialization, externalization, internalization and combination	Task characteristics, KM process, KMS capabilities	Fit	KM performance	Knowledge quality, knowledge satisfaction, creativity
Chen and Fong (2015)	Acquisition, dissemination, utilization and identification	Governance mechanism	KM process	Organizational performance	Knowledge satisfaction from Becerra-Fernandez and Sabherwal (2001)
Gold <i>et al.</i> (2001)	Acquisition, conversion, application and protection	Knowledge Infrastructure capability, knowledge process capability	N/A	Organizational effectiveness	KM performance index (KMI) with KM capability index, business performance index
Lee and Choi (2003)	Knowledge creation processes	Structure, culture, people, IT	Organizational creativity	Organizational performance	14 items (e.g. innovation, coordinating efforts and responsiveness)
Zheng <i>et al.</i> (2010)	Acquisition, conversion and application	Structure, culture, strategy	KM effectiveness	Organizational effectiveness	Market share, growth rate, profitability, innovation
Becerra-Fernandez and Sabherwal (2001)	Socialization, externalization, internalization and combination	KM processes	Task characteristics	KM effectiveness	Overall success, market share, profitability, growth rate, innovation
					11 items of KM satisfaction as KM effectiveness

(continued)

Table I	Previous studies	Focused KM processes	Independent variables	Mediators or moderators	Dependent variable	Measured items for dependent variable
	Sabherwal and Becerra-Fernandez (2003)	Socialization, externalization, internalization and combination	SEIC constructs	Mixed	KM effectiveness	11 items of KM satisfaction as KM effectiveness
	Chou <i>et al.</i> (2005)	Socialization, externalization, internalization and combination	Innovativeness, Inter-functional Coordination	SEIC constructs	KM satisfaction	4 items of KM satisfaction from Becerra-Fernandez and Sabherwal (2001)
	Lin (2007)	Acquisition, conversion, application and protection	N/A	N/A	KM effectiveness	9 items of KM satisfaction
	Song <i>et al.</i> (2007)	Codification and personalization	Knowledge processing orientation, network structure	Fit	KM effectiveness	3 items of KM satisfaction, adopted from Sabherwal and Becerra-Fernandez (2003)
	Tseng and Fan (2011)	Acquisition, storage, application and sharing	Organizational ethical climate, km attitude	KM process engagement	KM satisfaction	11 items of KM satisfaction from Becerra-Fernandez and Sabherwal (2001); 11 items of job performance
	Downes and Marchant (2016)	Creation, representation, transfer and application	KM process, process management	N/A	KM effectiveness	Collaboration, communication, learning, performance
	Masa'deh <i>et al.</i> (2017)	KM process –identification, creation, storage, application, etc.	KM process	KM satisfaction	Job performance	3 items of KM satisfaction adopted from Becerra-Fernandez and Sabherwal (2001)

and Paruchuri, 2005; Wang *et al.*, 2014; Hollenbeck and Jamieson, 2015). The more structural holes a unit spans, the fewer its redundant contacts (Ahuja, 2000; Burt, 1992; Burt, 2007). Burt (2000) argues that units that span more structural holes are likely to accumulate more social capital and enjoy better performance than those with fewer structural holes. Thus, such units afford benefits such as the creation of new knowledge and enhanced potential for innovative behavior (Obstfeld, 2005; Reagans *et al.*, 2004; Rodan, 2010; Phelps *et al.*, 2012).

From the social network perspective, there are two different views on how network positions create social capital. The traditional “closure” view emphasizes the positive effects of dense and cohesive ties on a normative environment that facilitate trust and cooperation between actors (Coleman, 1988, 1990), which in turn affords benefits such as exchange of information and knowledge (Reinholt *et al.*, 2011). The structural hole theory (Burt, 1992, 1997, 2007), on the other hand, claims that benefits from social capital result from brokerage opportunities, as actors who span structural holes are able to access diverse information.

These two perspectives lead to opposite predictions on how the inter-unit network structures may affect a unit's performance in task environment (Gargiulo and Benassi, 2000; Reagans and Zuckerman, 2001; Sosa *et al.*, 2015). According to the traditional “closure” perspective, in a closed network, social closure (i.e. lack of structural holes) fosters the development of trust and shared norms of behaviors, which in turn facilitates knowledge exchange and collaboration between units (Ahuja, 2000; Coleman, 1988; Tortoriello, 2015). Further, dense ties between units are likely to restrain opportunism (Ahuja, 2000; Coleman, 1988; Rowley *et al.*, 2000). Thus, highly interconnected network enables units to enjoy informational benefits, such as fine-grained information transfer and joint problem-solving (Ahuja, 2000; Reagans and Zuckerman, 2001; Uzzi, 1997; Perry-Smith and Shalley, 2014).

This study uses the network variable *constraint*, which refers to the extent to which a work unit's network is concentrated in redundant contacts (Burt, 1992). The *constraint* is a function of direct communication between units. Units that span more structural holes are less constrained. Greater benefits, such as access to diverse information, accrue to units that have lower *constraints* or more non-redundant contacts. On the contrary, network with fewer structural holes (i.e. more redundant contacts) may promote collective behaviors and reduce opportunism, which in turn may also enhance the unit's effectiveness in their use of knowledge. Thus, the following two hypotheses are supported by previous studies and theories:

- H2a. The extent of the redundant contacts (lack of structural holes) that a work unit maintains is positively associated with the work unit's KM effectiveness.
- H2b. The extent of the redundant contacts (lack of structural holes) that a work unit maintains is negatively associated with the work unit's KM effectiveness.

2.3 Individual work unit's task environment, knowledge management strategies and knowledge management effectiveness

Much of KM activities, such as knowledge creation, sharing and storage, occurs in individual work units, as an employee typically spends most of his or her working hours in the work unit interacting with colleagues or team members. In this section, we examine the relationships between task environment, KM strategies and KM effectiveness in the context of an individual work unit [Figure 1(b)].

In terms of KM strategies, a work unit can generally follow a codification strategy or a personalization strategy (Hansen *et al.*, 1999; Kim *et al.*, 2014) in its KM activities: creating, transferring and utilizing knowledge. While the codification strategy involves mostly formal documentation, the personalization strategy relies mainly on face-to-face interactions between workers (Gammelgaard and Ritter, 2005; Choi *et al.*, 2008; Cohen and Olsen, 2015). For example, procedures for handling accounts payable can be readily documented

(i.e. codified). In these situations, knowledge may be codified by means of manuals, formal documents, and databases, etc. Personalization, on the other hand, refers to how people interact and share their knowledge with others. For example, peculiar ways to handle certain types of customers would be difficult to describe and codify, and the sharing and transfer of such knowledge would benefit greatly from personal interactions between employees.

A task environment can be characterized by task variety and analyzability (Perrow, 1967; Byström, 2002; Lee *et al.*, 2015). Task variety refers to the number and frequency of exceptional, unexpected and/or novel events in the task over time (Daft and Macintosh, 1981; Karimi *et al.*, 2004; Wipawayangkool and Teng, 2016). According to the information processing theory, organizations process information to reduce uncertainty and equivocality (Daft and Lengel, 1986; Galbraith, 1977). Task variety represents uncertainty about the occurrence of problems and exceptions in performing a task (Daft and Macintosh, 1981; Chae *et al.*, 2015). Thus, when task variety is high, people have greater difficulty in predicting problems or activities in advance (Daft and Macintosh, 1981; McLain, 2009; Mehta *et al.*, 2014), and they would normally devote more efforts in seeking information/knowledge from formally recorded documents or computer files (the codification strategy) or by consulting colleagues through face-to-face contacts (the personalization strategy):

- H3a. In an organizational work unit, the level of task variety is positively associated with its reliance on the personalization KM strategy.
- H3b. In an organizational work unit, the level of task variety is positively associated with its reliance on the codification KM strategy.

Task analyzability refers to the extent to which a task can be broken into small, well-defined components (Perrow, 1967; Ahuja and Carley, 1999). Analyzable tasks involve more rules and procedures. Thus, when the environment for a work unit is permeated mostly by analyzable tasks, there are more information cues available and more procedures established (Rice, 1992; Karimi *et al.*, 2004; Lee *et al.*, 2015), and people are expected to process more formalized and written form of information. On the other hand, in a work unit with unanalyzable tasks, information processing is “more personal, less linear, more *ad hoc* and improvisational” (Daft and Weick, 1984, p. 287) than for units with more analyzable tasks. For unanalyzable tasks, people are required to seek satisfactory solutions to problems outside of their domains of rules and procedures (Rice, 1992; Oke and Idigbon-Oke, 2010; Choo *et al.*, 2013; Chae *et al.*, 2015). Further, people are more likely to use rich media, such as face-to-face communication, for accomplishing the task (Rice, 1992; Van de Ven *et al.*, 1976). Thus:

- H4a. In an organization work unit, the level of task analyzability is negatively associated with its reliance on the personalization KM strategy.
- H4b. In an organization work unit, the level of task analyzability is positively associated with its reliance on the codification KM strategy.

Through the internalization process (Nonaka, 1994), explicit knowledge may be embodied in action and practice, and thus, individuals are able to absorb it and turn it into tacit knowledge. Face-to-face communication (i.e. personalization) and learning by doing are examples of internalization processes by which individuals acquire knowledge. Sabherwal and Becerra-Fernandez (2003) found that internalization facilitates individual-level KM effectiveness. In addition, we may expect individuals to enhance effectiveness by exchanging tacit knowledge through joint activities. Thus, a work unit would have high KM effectiveness if the unit relies heavily on personalization in its KM activities (Storey and Kahn, 2010; Kushwaha and Rao, 2015):

- H5. In an organization work unit, the extent of its reliance on the personalization of KM strategy is positively associated with the degree of KM effectiveness.

To make tacit knowledge agreeable and understandable to others, it should be “objectivated” (von Krogh and Roos, 1995). Among the four types of knowledge creation

processes (Nonaka, 1994), externalization refers to the process of transforming tacit knowledge into documented forms that others can understand, which is central to the codification KM strategy (Hansen *et al.*, 1999; Kim *et al.*, 2014; Cohen and Olsen, 2015). Sabherwal and Becerra-Fernandez (2003) found that externalization facilitates individual-level KM effectiveness. Thus, a work unit is more likely to have higher KM effectiveness if members of the unit more actively engage in the codification activities in managing knowledge (Haas and Hansen, 2007; López-Nicolás and Meroño-Cerdán, 2011):

- H6.* In an organization work unit, the extent of its reliance on the codification KM strategy is positively associated with the degree of KM effectiveness.

3. Research methodology

The survey method was used to empirically test the hypotheses. We developed the items in the questionnaire by adapting previously validated instruments. We administered two survey forms: a member-survey form that obtained responses from members of each work unit and a leader-survey form issued only to the leader of each unit to obtain his/her perception on inter-unit communications. The sample frame consists of experienced knowledge workers, which typically includes managers and professionals. Drucker (1994) defined knowledge workers as high-level employees who apply theoretical and analytical knowledge, acquired through formal education, and who have a habit of continuous learning. Managers and professionals hold the typical characteristics of knowledge workers and were found to play key roles in KM (Choi and Lee, 2003; Janz *et al.*, 1997). A pilot test was undertaken with 12 experienced knowledge workers, which resulted in some refinement to the questionnaire.

The refined instrument was used to collect data from a large multi-unit advanced aerospace technology company located in a large southern city. The member-survey form was administered to a total of 500 managers and professionals in the company, and 156 responses were returned (31 per cent response rate). Additional questionnaires (i.e. the leader-survey) to measure inter-unit network structure were distributed to the team leaders of 27 work units, and 20 responses were returned (74 per cent response rate). Out of the 156 responses, 27 had incomplete data and were eliminated from further analysis. Among the remaining 129 responses, 84 were from these 20 work units. As a result, 20 leader responses and 84 member responses were used in the data analysis.

3.1 Sample characteristics

At the beginning of the questionnaire, we provide a definition of knowledge and explain what a “work unit” is. The respondent is also asked to indicate the name of the unit he/she belongs to and also briefly describe its activities. In addition, the questionnaire asks for basic demographic information from the respondents. The age of respondents ranged from 25 to 65, with an average of 48.04 (SD = 8.86). Consequently, knowledge workers of virtually all ages were represented in our sample. The presence of both genders in our sample also appears to be consistent with what one might expect in a population of knowledge workers: 68 per cent male and 32 per cent female. The mean number of years a respondent had spent with his/her present unit was 8.26 (SD = 7.54). Thus, we can be reasonably sure that they assessed the conditions in their units with a relatively high degree of accuracy. The average size of work units was 27.1 people (SD = 27.12; 32 for 75 percentile). About one-third (35.2 per cent) of the respondents classified themselves as managers, and the remainder consisted of engineers, analysts, specialists and a variety of other knowledge workers. These sample characteristics provide strong indications that the respondents are reasonably representative of the population of experienced knowledge workers.

3.2 Measurement of research variables

KM effectiveness. The items for KM effectiveness were developed based on our conceptualization of KM effectiveness discussed earlier (Appendix 1). This encompasses satisfaction with the stages of the KM cycle, i.e. the process of knowledge creation, acquisition, storage, transfer and application. A seven-point Likert scale was used, anchored by “strongly disagree” (1) and “strongly agree” (7).

Network structure. On the leader-survey form, the names of all the work units involved in the study were listed. We assessed work unit relations by asking team leaders a question: “In the past year, how much information did you obtain on work-related matters from this team?” (Burt, 1992; Ibarra, 1993; Sparrowe et al., 2001). A five-point Likert scale was used, anchored by “none” (1), “some” (3) and “very much” (5). This scale is the basis of the two network structural measures described below.

We considered *indegree* centrality and *network constraints* (lack of structural holes). *Indegree* is the most suitable centrality measure for capturing an actor’s information or knowledge access (Freeman, 1979; Tsai, 2001). The sum of *network constraints* is considered as a lack of structural holes (Gargiulo and Benassi, 2000).

Indegree centrality. *Indegree* centrality refers to a work unit’s power or status in a network (Wasserman and Faust, 1994). In addition, *indegree* indicates the extent to which a work unit serves as a knowledge source or advisor to other units (Hansen, 2002). *Indegree* centrality of a work unit n_i was computed as:

$$C(n_i) = \frac{x_{+i}}{g-1}$$

where x_{+i} represents the number of ties received by work unit i and $g-1$ is the maximum possible number of ties linking i to other units (Wasserman and Faust, 1994). We used a constraint-based approach to measuring structural holes (Gargiulo and Benassi, 2000; Nerkar and Paruchuri, 2005).

Network constraint refers to the extent to which a work unit’s network is concentrated in redundant contacts (Burt, 1992). They were computed following Freeman’s (1979) definition in UCINET 6 software package (Borgatti et al., 2002). *Network constraint* was measured as:

$$C_{ij} = (p_{ij} + \sum_q p_{iq} * p_{qj})^2$$

For $q \neq i, j$ where p_{ij} is the proportion of work unit i ’s relations invested in contact j , and $\sum_q p_{iq} * p_{qj}$ is the portion of work unit i ’s relations invested in contacts q who are in turn invested in contact j (Burt, 1997).

KM strategies: personalization and codification. The questionnaire items were developed by Song et al. (2007). The measures reflected the extent of relying on formal documentation or spontaneous face-to-face interaction throughout the KM cycle – creation, acquisition, sharing and transfer and utilization. We used the seven-point Likert scale, anchored by “to a very little extent” (1) and “to a very great extent” (7).

Task characteristics: task analyzability and task variety. Both scales were developed and validated by Daft and Macintosh (1981), who reported a Cronbach’s coefficient alpha = 0.77 for task variety and 0.86 for task analyzability. We used the seven-point Likert scale, anchored by “to a very little extent” (1) and “to a very great extent” (7) (Table II).

3.3 Reliability and validity

For the intra-unit analysis, given the size of the sample ($N = 84$), we tested the research model using the partial least square (PLS) method with SmartPLS 2.0M (Ringle et al., 2005).

Before testing the various hypotheses, we first examined the reliability and validity of the scales. The composite reliability (CR) values are reported in Table III. If CR values are less than 0.70, the items may be unrelated, or measuring more than one construct. As seen in the table, all CR values are above 0.70 (range from 0.847 to 0.947), and thus deemed acceptable (Fornell and Larcker, 1981). To ascertain the discriminant and convergent validity, we require the average variance extracted (AVE) to exceed 0.50, or the square root of AVE to be greater than the correlation between a construct and any other construct (Chin, 1998). As can be seen in Table III, the square root of the AVEs (on the diagonal) is indeed greater than the corresponding correlations. We have also carried out a principal component factor analysis with varimax rotation. The results, as indicated in Table IV, reveal that all the constructs are clearly delineated.

3.4 Levels of analysis

The unit of analysis is individual for intra-unit analysis (H3-H6). All task-related and KM-related measures were measured at the individual level. For inter-unit analysis (H1 and H2), the unit of analysis is at the work unit level. Network structure, as described earlier, was measured at the work unit level by collecting responses from unit leaders. Individual perceptions on KM effectiveness were aggregated by taking the average of unit members' responses and expressing them as the work unit value (Janz et al., 1997). The long tenure of most work unit members indicated that our perceived measures reflected a view of the collective (Hackman, 1992; Janz et al., 1997).

To justify using the average value of work unit members as an indicator, it was critical to demonstrate high within-work unit agreement. The inter-rater agreement represents interchangeability among raters (James et al., 1984). Agreement was assessed using the $r_{wg(j)}$ index. The median value of $r_{wg(j)}$ for KM effectiveness was 0.93. George (1990) indicated that aggregation is justified by a median $r_{wg(j)}$ value of 0.70 or greater. The values of $r_{wg(j)}$ were also consistent with the findings of other studies (Choi and Lee, 2003; Janz et al., 1997).

Table II Descriptive and correlations of research variables: inter-units ($N = 20$)

Constructs	Means	SD	Correlations	
			1	2
1. KM effectiveness ^a	4.18	0.96		
2. Degree centrality ^b	0.29	0.10	0.39	
3. Structural holes ^c	0.23	0.02	−0.51	−0.18

Notes: ^aThe measure was aggregated by work unit; ^bDegree centrality was measured by using indegree centrality; ^cStructural holes were measured by using a network constraint-based approach

Table III Descriptive, CRs, correlations and AVE values: intra-unit ($N = 84$)

Constructs	Means	SD	CR	AVE	Correlations				
					1	2	3	4	5
1. KM effectiveness	4.31	1.17	0.944	0.773	<i>0.879</i>				
2. Codification	4.39	1.21	0.936	0.784	0.586**	<i>0.886</i>			
3. Personalization	5.39	0.85	0.907	0.709	0.281**	0.068	<i>0.842</i>		
4. Task variety	4.80	0.92	0.847	0.582	0.189	0.274*	0.390**	<i>0.763</i>	
5. Task analyzability	4.86	1.15	0.905	0.704	0.438**	0.480**	−0.059	−0.022	<i>0.839</i>

Notes: CR: composite reliability; AVE: average variance extracted; italic numbers on the diagonal are the square roots of the AVE values; *indicates $p < 0.05$ and **indicates $p < 0.01$

Table IV Exploratory factor loading for research variables (intra-unit, $N = 84$)

	<i>Components</i>				
	1	2	3	4	5
KSAT1	0.874	0.077	−0.039	0.115	0.089
KSAT2	0.902	0.165	0.117	0.130	0.088
KSAT3	0.674	0.388	0.298	0.165	0.027
KSAT4	0.748	0.363	0.248	0.143	−0.012
KSAT5	0.819	0.328	0.134	0.285	0.047
EO1	0.193	0.712	−0.085	0.378	0.249
EO2	0.237	0.859	0.026	0.081	0.030
EO3	0.232	0.827	0.072	0.250	0.118
EO4	0.377	0.774	−0.028	0.196	0.177
TO1	0.033	0.084	0.855	−0.101	0.105
TO2	0.170	−0.061	0.865	0.017	0.083
TO3	0.100	0.027	0.804	−0.014	0.151
TO4	0.106	−0.013	0.730	−0.029	0.315
TV1	0.058	0.214	0.126	−0.037	0.772
TV2	−0.002	0.229	0.009	−0.168	0.859
TV3	0.147	−0.196	0.286	0.031	0.663
TV4	0.012	0.112	0.278	0.049	0.606
TA1	0.085	0.240	0.029	0.778	−0.145
TA2	0.055	0.106	0.062	0.847	−0.062
TA3	0.325	0.106	−0.125	0.801	0.210
TA4	0.267	0.239	−0.149	0.760	−0.099

Note: "Italic numbers are factor loadings on corresponding factors"

3.5 Hypotheses testing

We used regression analysis to examine the relationship between inter-unit network structure and KM effectiveness. Results of the regression analysis are summarized in Table V.

The results showed that a work unit occupying a more central position is likely to have higher KM effectiveness, confirming $H1$ ($b = 0.54$, $t = 1.76$, $p < 0.05$). As hypothesized in $H2b$, there was a significant negative relationship between the redundant contacts that a work unit maintains and its KM effectiveness ($b = -3.39$, $t = 2.46$, $p < 0.05$).

Results of the PLS analysis for intra-unit, including all R^2 and path coefficients values, are reported in Table VI and also indicated in Figure 2. R^2 values indicate the amount of variance explained by the model (Chin, 1998). The model explained 31.7 per cent of the variance in the use of codification KM strategy, 18.0 per cent of variance in the use of personalization strategy and 42.3 per cent of variance in KM effectiveness.

The results show that task variety is positively associated with personalization KM strategy usage, providing support for $H3a$ ($b = 0.418$, $t = 4.92$, $p < 0.001$). Task variety also has a significant relationship with the work unit's codification strategy usage ($H3b$: $b = 0.274$, $t = 3.28$, $p < 0.001$). As expected, task analyzability is significantly related to the use of

Table V Effects of inter-unit network structure on KM effectiveness ($N = 20$)

<i>Constructs</i>	R^2	β	<i>t-value</i>
<i>Dependent: KM effectiveness</i>			
Degree centrality ^a	0.15	0.54	1.76*
Structural holes ^b	0.26	−3.39	2.46*

Notes: *indicates $p < 0.05$; **indicates $p < 0.01$; ^aDegree centrality was measured by using *indegree centrality*; ^bStructural holes were measured by using a *network constraint*-based approach

codification strategy, and we found support for *H4b* ($b = 0.499$, $t = 5.89$, $p < 0.001$). However, there was no relationship between task analyzability and the personalization strategy, thus no support for *H4a* ($b = -0.061$, $t = 0.53$).

As hypothesized, use of the personalization strategy is significantly related to KM effectiveness, supporting *H5* ($b = 0.252$, $t = 2.94$, $p < 0.01$). Finally, our research also found support for the positive association between the codification strategy usage and KM effectiveness, supporting *H6* ($b = 0.582$, $t = 8.94$, $p < 0.001$).

4. Discussions and conclusions

We have proposed two research questions to examine the relationship between inter-unit network structures and KM effectiveness, as well as a set of four hypotheses to investigate the effects of intra-unit task environment on KM effectiveness, through different KM strategies. The results of testing hypotheses are summarized in Table VII.

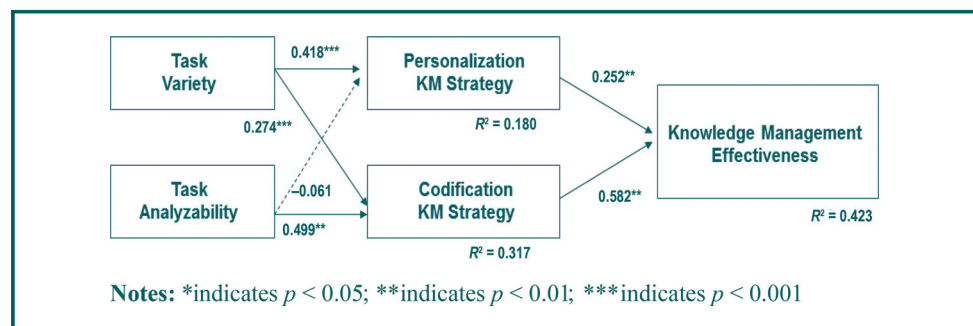
As hypothesized, units that have a more central position are likely to have higher KM effectiveness (*H1* supported). High degree of centrality in a network that is characterized by

Table VI Results of PLS analysis: intra-unit, $N = 84$

	<i>Personalization</i>	<i>Codification</i>	<i>KM effectiveness</i>
R^2	0.180	0.317	0.423
<i>H3a and H3b</i>			
Task variety			
Path coefficient	0.418***	0.274***	
<i>t</i> -value	4.92	3.28	
<i>H4a and H4b</i>			
Task Analyzability			
Path coefficient	-0.061	0.499***	
<i>t</i> -value	0.53	5.89	
<i>H5</i>			
Personalization			
Path coefficient			0.252**
<i>t</i> -value			2.94
<i>H6</i>			
Codification			
Path coefficient			0.582***
<i>t</i> -value			8.94

Notes: *indicates $p < 0.05$; **indicates $p < 0.01$; and ***indicates $p < 0.001$

Figure 2 Testing research Model II (intra-unit)



high communication and knowledge exchange activities enables units to access useful knowledge from other units (Tsai, 2001). Thus, these units are expected to enjoy informational benefits, which in turn enhance their effective use of knowledge. KM effectiveness is negatively impacted by the number of redundant contacts a unit maintains (*H2b* supported), which indicates that structural holes spanned by a unit provide opportunities for the unit to broker and control the flow of information between units. Further, structural holes enable the unit to construct an efficient, information-rich network with other units, in which the unit can afford the resource sharing benefits (Burt, 1992; Ahuja, 2000; Zaheer and Bell, 2005). Our results are consistent with previous studies (Tsai, 2000; Reinholt *et al.*, 2011; Liao and Phan, 2016) on social networks that demonstrate that network positions at work are important contributors to KM effectiveness.

From the intra-unit analysis, our results showed that task variety is significantly related to the use of both personalization and codification KM strategies in the unit (*H3a* and *H3b* supported). Interestingly, the link to personalization (i.e. path coefficient = 0.418) appears to be much stronger than that for codification (i.e. path coefficient = 0.274). This seems to indicate that variety in the task environment would make it more unstructured and thus engender more need for personalization than codification. Future studies may further explore the theoretical and practical dimensions of this comparison.

As seen in Figure 2, task analyzability has a significant influence on the codification strategy usage (*H4b* supported), but not on the personalization strategy usage (*H4a* not supported). This result means that in a work unit with analyzable tasks, people are most likely to process information or knowledge in a formalized and written form (Rice, 1992; Karimi *et al.*, 2004). Interestingly, we found partial support for our argument that task analyzability is negatively related to the use of personalization strategy. This finding indicates that information processing with rich media such as face-to-face communication may be required for unanalyzable tasks.

Both types of KM strategies have positive impacts on KM effectiveness (*H5* and *H6* supported). These findings suggest that KM at the work unit level needs to maintain a balance between the personalization and codification strategies. However, we also see evidence that this strategy is contingent upon the unit's task environment, and that there is no universally appropriate KM strategy for a unit.

Table VII Summary of testing hypotheses

	<i>Hypothesized path</i>	<i>b</i>	<i>Std. path coefficient</i>	<i>t-value</i>	<i>Hypothesis support</i>
<i>H1</i>	Centrality of a work unit's network position → KM effectiveness	0.54		1.76*	Yes
<i>H2a</i>	Lack of structural holes → (+) KM effectiveness	−3.39		2.46*	No
<i>H2b</i>	Lack of structural holes → (−) KM effectiveness	−3.39		2.46*	Yes
<i>H3a</i>	Task variety → (+) personalization KM strategy		0.418	4.92***	Yes
<i>H3b</i>	Task variety → (+) codification KM strategy		0.274	3.28***	Yes
<i>H4a</i>	Task analyzability → (−) personalization KM strategy		−0.061	0.53	No
<i>H4b</i>	Task analyzability → (+) codification KM strategy		0.499	5.89***	Yes
<i>H5</i>	Personalization KM strategy → (+) KM effectiveness		0.252	2.94**	Yes
<i>H6</i>	Codification KM strategy → (+) KM effectiveness		0.582	8.94***	Yes

Notes: *indicates $p < 0.05$; **indicates $p < 0.01$; and ***indicates $p < 0.001$

4.1 Conclusion and implications

The current study attempts to provide an understanding of the influence of inter-unit social network structure on KM effectiveness and the influence of intra-unit task environment and KM strategies on KM effectiveness. Our results have significant implications for KM research. First, previous KM studies on the impact of network characteristics of intra-organizational units have been limited in scope, focusing primarily on knowledge sharing (Hansen, 2002; Hansen *et al.*, 2005; Nakauchi *et al.*, 2017). Our research examines the impacts of network structures on overall KM effectiveness, which encompasses the entire cycle of KM activities – including knowledge creation, sharing and utilization – rather than a partial portrait about particular KM activities. Further, deeper insights may be gained by our study of relating multiple measures (i.e. degree centrality and structural holes) of network structure to overall KM effectiveness. Second, the results reveal the relative strength of a unit's network position in its effective use of knowledge. This finding resonates with the findings of Reagans and Zuckerman (2001), who demonstrated that team productivity is positively influenced by demographic diversity and the strength of the relationship between actors (Han *et al.*, 2014). Thus, future studies may look at the antecedents and facilitators to explicate the relationship between two network perspectives (i.e. closure and structural hole views) and KM effectiveness. Third, according to Te'eni's (2001) research, task variety is one of the most significant factors that can potentially explain the complex phenomena in organizational communications. Our findings are also consistent with prior studies that the effectiveness of KM processes may vary depending upon work units' task characteristics (Lindsey, 2002; Chang *et al.*, 2012; Downes and Marchant, 2016). Thus, further consideration of how an understanding of different task environments in different work units can be leveraged to improve KM effectiveness merits discussion. Our results have shed more light into this emerging line of inquiry in the context of KM practices. Finally, previous studies on KM strategies (Kim *et al.*, 2014) primarily focus on the direct influence of KM strategy on organizational effectiveness. In contrast, we found that a unit's KM strategy plays a mediating role between intra-unit task characteristics and KM effectiveness, which leads to expand our understandings about the critical role of KM strategy to leverage various KM activities. Further, although our results showed that both personalization and codification strategies significantly relate to KM effectiveness, future research may explore the appropriate mixture of personalization and codification strategies under different task environments to improve knowledge work performance for daily KM processes and activities.

This study offers practical guidance to KM managers on not only how to exploit different network structures of a work unit to enhance KM effectiveness but also which KM strategies may fit to the work unit's task environments. This study also can be used as a tool to identify a work unit's current state in network structures and task characteristics along with its KM processes. First, our findings suggest that inter-unit knowledge exchange and a unit's brokering role are vital components of a work unit's KM strategy. The unit manager may enhance KM effectiveness by paying attention to its boundary-spanning communications via structural holes, as well as broadly encouraging interaction and communications with other units. Second, the work unit manager may improve the scope and effectiveness of the unit's KM practices by striking a balance between the personalization and the codification strategies for KM. In sum, for successful KM in a work unit, its manager should not only establish appropriate inter-unit relationships but also develop the unit's own KM strategies.

4.2 Future perspectives and limitations

We found interesting influence patterns of the relationships between inter-unit network structure and KM effectiveness. Further, our study has revealed the significant role of task environments in the intra-unit KM activities. Our study focuses attention on work-unit KM and specifically considers the influence of task characteristics in the unit. Our results have provided a basis for future inquiry, which may, for example, examine the relationships between KM strategies at the unit level and organizational level. Although *indegree* is

considered as the most suitable centrality measure (Freeman, 1979; Tsai, 2001), future studies need to consider other network measures, such as tie strength and network heterogeneity, to explore the relationship between network structure and KM effectiveness. In addition, other facilitators and antecedents, such as culture, trust and absorptive capacity, could be included to provide a richer understanding of the intra-organizational KM activities. For further research on KM effectiveness, more comprehensive measures may be required to capture the richness of this construct. Here, we have only utilized the theoretical perspective of network structures and task environments to examine KM effectiveness. Researchers can extend the theoretical perspective used in this study to investigate other outcome variables such as KM performance and organizational effectiveness.

We have uncovered a pattern of influence on KM effectiveness that has not been previously examined. These results, however, should be interpreted with caution. For the inter-unit analysis, we draw a sample of 20 work units from one organization. While our preliminary findings hold great promise, generalization of the results to work units universally requires further inquiry. For the intra-unit analysis, the results are based on individual responses. As in many other published studies, collecting all measures from a single respondent raises the concern of common method bias. In the case of this study, we have examined the common method variance, referring to variance resulting from the use of a common method rather than from the construct itself (Podsakoff *et al.*, 2003). We conducted Harman's single-factor test (Podsakoff *et al.*, 2003). In this study, all 21 items were analyzed by using principal component factor analysis. We found five factors, in which the first factor accounted for 33.28 per cent of the variance. Further, no general factor was apparent in the unrotated factor solution. The results indicated that common method variance was not a major problem in this study (Sabherwal and Becerra-Fernandez, 2003; Scott and Bruce, 1994). In addition, for independent variables such as task characteristics and KM strategies, the questions are targeted specifically at actual activities within a unit, with little or no filtering through individual perception. Thus, we conclude that common method bias has minimum impact on the validity of our results.

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Appendix 1. Instruments

All items were measured on a seven-point Likert scale. We provide a definition of knowledge and make a distinction between information and knowledge in the survey:

KM effectiveness (based upon [Sabherwal and Becerra-Fernandez, 2003](#)).

1: strongly disagree, 7: strongly agree.

Note: KM includes how knowledge is created, acquired, stored, shared and applied.

- KSAT1 I am satisfied with how new knowledge is created in my work unit.
- KSAT2 I am satisfied with how knowledge is acquired in my work unit.
- KSAT3 I am satisfied with how knowledge is shared and transferred in my work unit.
- KSAT4 I am satisfied with how knowledge is applied and utilized in my work unit.
- KSAT5 Overall, I am satisfied with knowledge management in my work unit.

KM strategy (adopted from [Song et al., 2007](#)).

1: to a very little extent; 7: to a very great extent.

Note: Formal documents include traditional papers, manuals, computer files, databases, repositories, Web (internets and intranets), etc. for explicitly recording knowledge.

Codification strategy

- EO1 Using formal documents to capture and describe knowledge.
- EO2 Formally recording knowledge whenever it is created (e.g. from projects and meetings).
- EO3 Using formal documents to share and transfer knowledge.
- EO4 Using knowledge and procedures from formal documents to solve problems.

Personalization strategy

- TO1 Making face-to-face social interactions to exchange knowledge.
- TO2 Engaging in informal dialogues and formal meetings to share and transfer knowledge.
- TO3 Using meetings and discussion via brainstorming and debate, etc. to generate new knowledge.
- TO4 Using knowledge from accumulated experience to solve problems.

Task variety (adopted from [Daft and Macintosh, 1981](#)).

1: to a very little extent; 7: to a very great extent:

- TV1 Work decisions are dissimilar from one day to the next.
- TV2 The work could be described as routine. (reverse-coded).
- TV3 When a problem arises, it takes a lot of experience and training to know what to do.
- TV4 There is variety in the events that cause the work.

Task analyzability (adopted from [Daft and Macintosh, 1981](#)).

1: to a very little extent; 7: to a very great extent:

- TA1 Normal work activities in our jobs are guided by standard directives, rules and procedures.

- TA2 In carrying out our work, there is an understandable sequence of steps that can be followed.
- TA3 Established materials (professional books, directives, manuals, statutes) cover our work.
- TA4 In our type of work, people actually rely on established practices and procedures.

Appendix 2

Table A1 Basis of social capital from a work unit's network positions			
Network position	Related network measures	Role of work units	Basis of social capital from network positions
Position of brokerage	Structural hole	when a unit serves as an <i>exclusive passage</i> of information flows among other units in the network	To control the flow of communications <i>KM-related benefits:</i> Access to more skills, expertise, and resources To enhance the unit's ability of absorbing knowledge Effective when requiring diverse information or expertise
Position of proximity	Closeness centrality	When a unit can reach other units <i>more directly</i> in the network	To quickly access diverse information <i>KM-related benefits:</i> To access complementary skills To conduct efficient search and effective transfer of knowledge Effective in knowledge transfer and utilization
Position of prominence	Degree centrality	When a unit has <i>high frequency of information flow</i> with other units in the network	To be a hub of communication and information exchange <i>KM-related benefits:</i> Opportunities to create new knowledge and to transfer and exchange tacit knowledge Effective in knowledge creation and exchange

Notes: The work units that comprise an organizational network structure vary in their degree of centralities, which may make some units become more prominent sources of social capital than others (Hansen, 2002; Song *et al.*, 2007). This table summarizes three different types of a work unit's network positions and the basis of its social capital

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